Exp No:16

SERIAL INTERRUPT USING UART (KEYBOARD INPUT) IN KEIL

PROGRAM:

```
#include &lt;reg51.h&gt;
#include &lt;stdio.h&gt;

void serial_ISR(void) interrupt 4
{
char ch;
if (RI)
{
RI = 0; // Clear receive interrupt flag ch =
SBUF; // Read received character

printf(&quot;\n&gt;&gt;&gt; INTERRUPT RECEIVED: &quot;);
printf(&quot;%c\n&quot;, ch);
}
}

void delay(void)
{

unsigned int i;
for(i = 0; i &lt; 50000; i++); // Delay for visibility }

void main(void)
{
SCON = 0x50; // UART mode 1, 8-bit, REN enabled
TMOD = 0x20; // Timer1 mode 2 (auto-reload) TH1 =
0xFD; // Baud rate 9600
TR1 = 1; // Start Timer1
```

IE = 0x90; // EA = 1, ES = 1 (enable serial interrupt) TI

= 1; // Enable printf

while(1)

{

printf("Hello World\n");

delay();

}

}

OUTPUT: